Using Computer Vision to Assist the Visually Impaired

A Multi-Modal Deep Learning Approach

Nawel Zait & Victor Micha

The Problem

253 Million People Worldwide Face Daily Challenges

Traditional aids (white canes) only detect immediate obstacles

- Cannot read street signs
- Cannot identify approaching vehicles
- No contextual understanding of scenes

Our Goal: Translate visual information into actionable audio guidance in real-time

Our Solution - Four Integrated Modules

Multi-Modal Deep Learning System

1. **YOLOv8** - Real-time obstacle detection & distance estimation
2. **EasyOCR** - Text recognition (street signs, labels)
3. **BLIP** - Natural language scene descriptions
4. **ImageNet Classifiers** - Product identification

Key Innovation: Interactive Q&A capability for natural language queries

System Architecture

Parallel Processing Pipeline

- Input: Camera video/images
- Four modules process simultaneously
- Output: Prioritized audio alerts + Interactive Q&A

Performance:

- 12.7 FPS real-time processing
- 85% distance estimation accuracy
- 100% environment detection accuracy

Module I - Object Detection (YOLO)

Real-time Visual Perception

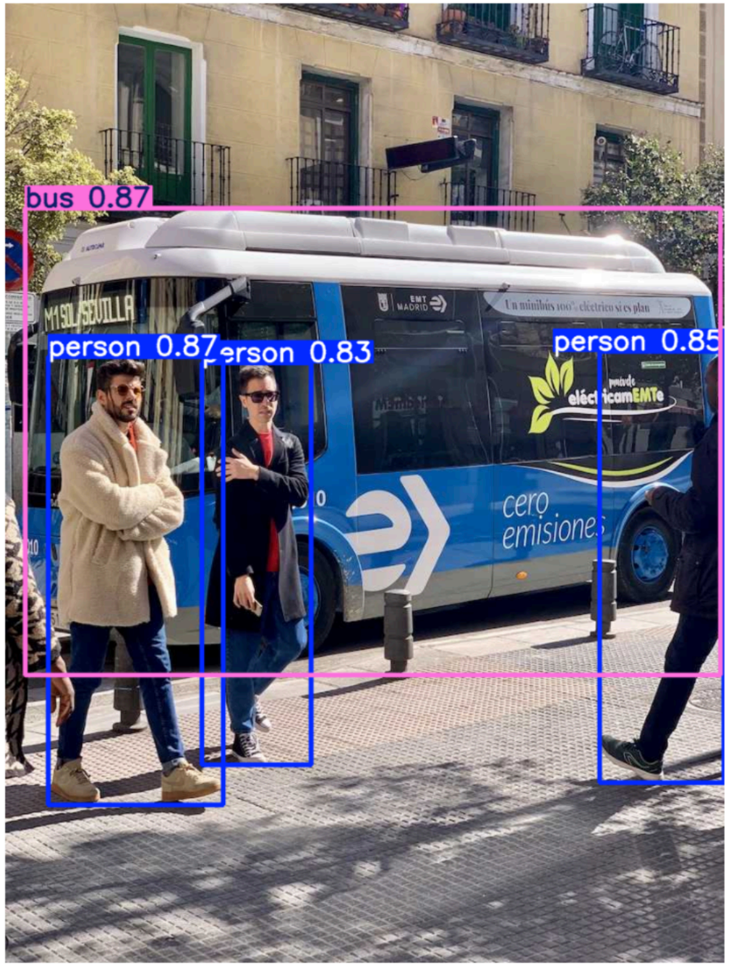
- YOLOv8n model (lightweight, fast)
- 80 object classes from COCO dataset
- 15.6 FPS on high-resolution video

Key Features:

- Distance estimation: Very Close (<2m), Close (2-5m), Far (>5m)
- Spatial positioning: Left/Center/Right
- Critical obstacle alerts: People, vehicles, traffic signs

Accuracy: 94% position accuracy, 85%

Détection d'objets - 4 objets détectés



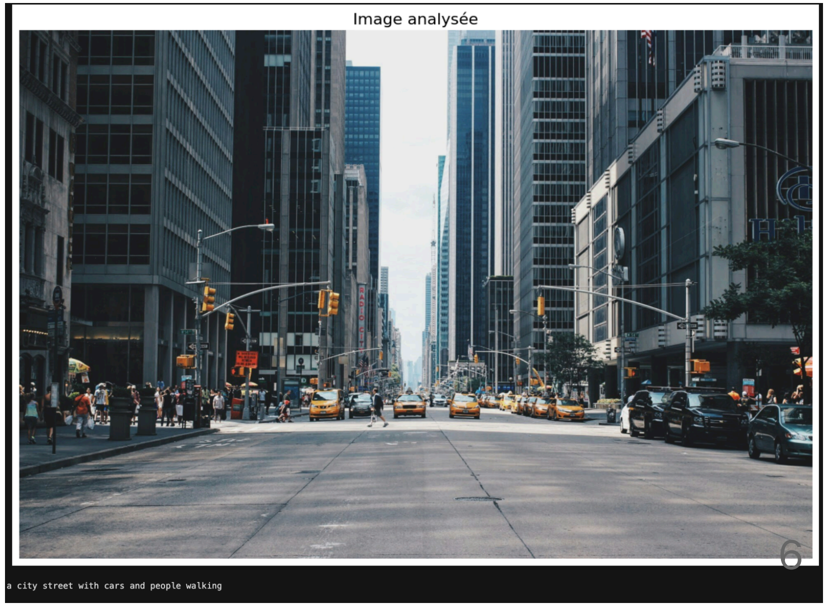
Modules II & III - Text & Scene Understanding

Module II: OCR Text Recognition

- Dual-engine: EasyOCR + Tesseract
- Reads street signs, license plates, building numbers
- Smart deduplication (85% similarity threshold)
- French/English support

Module III: Scene Description (BLIP)

- Generates natural language descriptions
- Updates every 3 seconds
- Auto-detects indoor/outdoor



Intelligent Alert System

Three-Level Priority System

1. **Critical** - Very close obstacles (<2m) - Immediate alerts
2. **Warning** - Nearby objects (2-5m) - Rate-limited
3. **Informational** - Street signs, scene descriptions - Lower frequency

Cognitive Load Management:

- Maximum 2 alerts/second
- 3-second minimum between similar alerts
- Smart deduplication to reduce redundancy

Interactive Q&A - Game Changer

Natural Language Queries:

- "Where am I?" → Identifies street names from OCR
- "What's around me?" → Lists detected objects
- "Is it safe?" → Evaluates obstacle distances
- "How many cars?" → Counts specific objects

Context-Aware Responses:

- Synthesizes information from all modules
- Adapts to indoor/outdoor environments
- 100% accuracy on spatial queries
- 83% accuracy on safety assessments

Results & Comparison

Our System vs. Prior Work (Ahmad et al.)

Advantages:

✓ OCR integration (absent in prior work)

✓ Interactive Q&A system (major innovation)

✓ Intelligent 3-level TTS alerts

✓ No custom training required - all pre-trained models

✓ True real-time capability (12.7 FPS)

Deployment Benefits:

- Immediate real-world deployment
- Works across diverse environments
- Comprehensive text support

Impact & Future Work

Real-World Impact

- Improved mobility and independence
- Real-time comprehensive environmental information
- Natural interactive dialogue capabilities

Future Enhancements:

- Dedicated depth model for precise distance (KITTI dataset)
- 30 FPS optimization for live video
- Multi-language support (Arabic, Spanish, Chinese)
- GPS integration for turn-by-turn navigation
- Edge deployment (mobile devices, Raspberry Pi)